System Validation Report

SE Miniproject | Form upload 2 of 3 | Team 10

Record	Verified value
Project	Collaborative Study Material & Quiz Platform (Study Commons)
Prepared by	Harshit Chandak - PES2UG24CS185; whole Team 10 evidence
Execution date	2026-11-05 (Asia/Calcutta)
PDF preparation date	2026-11-06; packaging existing results, not a new test run
Tested source	888c0666ae78e14be134a6c090be860d254ddb00
Integrated main commit	1c6ecafc3807a1f3916d79ea81416cd580e12eec (PR #44, merged and closed)
Preserved baseline	Submitted SRS, SAD and Test Plan unchanged

Scope covers Dharani (Auth/Materials/Dashboard), Prajwal (Question Bank/Quiz), Karan (Attempt/Result/History/Performance), and Harshit (Admin/Security/CI/shared integration).

Evidence links in this PDF point to immutable files in the integrated repository. Dates, environment, coverage limitations and outstanding acceptance items are retained; no approval is inferred from automated passes.

Execution date: **2026-11-05**, Asia/Calcutta. Source: `888c0666ae78e14be134a6c090be860d254ddb00`. Branch: `PES2UG24CS185-Sprint-2`. The execution manifest identifies the tested source and reporting configuration.

Scope: **the whole Team 10 project**, covering Auth/Materials (Dharani), Question Bank/Quiz (Prajwal), Attempt/Result/History/Performance (Karan), Admin/Security/CI and shared integration (Harshit). Execution coordination does not replace independent peer acceptance.

Test level	Passed	Failed	Skipped
backend integration	131	0	0
backend unit	199	0	0
browser system	21	0	0
frontend component/service/integration	190	0	0

Executed validation

- Full backend typecheck, production build, regression and coverage passed. Frontend complete regression, lint and compiled local release build passed.
- Complete Chrome/Edge/Firefox browser matrix: 21 verified cases; individual journeys and channels are listed in the case catalogue. Includes role navigation, PDF upload/search/view/download/removal, Question Bank/Quiz creation and discovery, trusted persisted scoring, result/history/performance, theme/responsive behavior and confirmed bulk moderation including hidden items. These are local loopback/disposable-database results.
- Built release smoke and actual database/application stop/restart passed (`demo:verify`): compiled assets, authentication/RBAC, PDFs, saved attempts, idempotency, result/download contracts, removal and historical data retention. Saved demo data and Atlas were not used for destructive testing.
- Submitted-document fingerprint guard passed. SRS, SAD and Test Plan were preserved.

Browser versions observed for the installed channels immediately after the matrix:

Browser channel	Version
chrome	156.0.8078.80
edge	156.0.4314.58
firefox	157.0

Acceptance still to execute or document, 2-6 November

Test Plan / SRS item	Current evidence	Remaining evidence / owner
NF-001 normal-network timing	Automated local content-ready assertions under 3 seconds	Team records typical campus/broadband device/network measurements
NF-003 concurrent normal use	10 isolated browser contexts concurrently complete login, PDF upload, Quiz creation, saved submission, owner history and performance in each of Chrome/Edge/Firefox; 10/50 HTTP-session workloads also pass	Automated isolated-browser-session scenario passed; independent peer observation remains separate
NF-004 latest two stable major versions	One installed Chrome/Edge/Firefox channel each, actual versions above	Team supplies acceptance-time latest/previous-major version matrix and results; three engines alone do not satisfy six version slots
NF-005 usability	Automated navigation, error and responsive assertions	Representative user completes core workflow without assistance; record observation
SR-005 deployed HTTPS	Production transport tests locally; all five jobs passed in exact-source CI run 41988979219, including the certificate-validated HTTPS container smoke	Exact-source HTTPS automated evidence is available in CI; no public deployment claim and no local Docker execution
Independent peer cross-testing	Automated checks do not establish an independent human review	Prajwal tests Auth/Materials; Karan tests Question/Quiz and Admin/Security; Dharani tests Attempt/Result/History/Performance; all test integration
UAT / exit / release	No automated failures in executed cases	QA/team/Guide review, complete RTM/defect log, genuine acceptance, demo and release freeze

Defects and conclusion

The final automated run passed without failed or skipped cases. Historical failures remain recorded in their original dated packs; confirmed audit findings are listed below when applicable. Lower frontend coverage and incomplete acceptance evidence are explicitly retained as gaps. The historical CSMQ-45 latest-50/weighted-average description was corrected in the supplementary matrix to match the current all-pages/arithmetic-mean implementation. Generated coverage/build files were excluded from frontend lint inputs.

Automated validation passed; complete Test Plan/UAT exit is still pending the evidence above. No human approval, unexecuted browser version or unexecuted test result is fabricated. See updated cases and case catalogue.

Source CI: exact-source CI run 41988979219, all five jobs successful (backend, three browsers and containers). The tested application source was subsequently merged into main by PR #44. This PDF packages the completed 5 November execution; no new 6 November test execution is implied.

Concurrent browser-session evidence

Channel	Actual version	Sessions	Completed	Page errors	Elapsed ms
chrome	156.0.8078.80	10	10	0	37058
edge	156.0.4314.58	10	10	0	38609
firefox	157.0	10	10	0	51149

Single-user primary-content timings

Local same-origin fixture, application already started; browser actions/content-ready assertions, 3,000 ms limit. Concurrency timings above are a separate functional workload.

Channel	Page/action	Milliseconds	Limit
chrome	Materials	250	3000
chrome	Create Quiz	373	3000
chrome	Quiz Search	218	3000
chrome	Student Dashboard	368	3000
chrome	Quiz View	207	3000
chrome	Quiz Attempt	325	3000
chrome	Submission and Result	590	3000
chrome	Attempt History	184	3000
chrome	Quiz View	358	3000
chrome	Quiz Attempt	180	3000
chrome	Submission and Result	429	3000
chrome	Attempt History	351	3000
chrome	Subject Performance	303	3000
chrome	Admin Dashboard	207	3000
edge	Materials	188	3000
edge	Create Quiz	276	3000
edge	Quiz Search	162	3000
edge	Student Dashboard	146	3000
edge	Quiz View	141	3000
edge	Quiz Attempt	126	3000
edge	Submission and Result	324	3000
edge	Attempt History	169	3000
edge	Quiz View	213	3000
edge	Quiz Attempt	169	3000
edge	Submission and Result	360	3000
edge	Attempt History	118	3000
edge	Subject Performance	238	3000

Channel	Page/action	Milliseconds	Limit
edge	Admin Dashboard	350	3000
firefox	Materials	294	3000
firefox	Create Quiz	275	3000
firefox	Quiz Search	326	3000
firefox	Student Dashboard	204	3000
firefox	Quiz View	250	3000
firefox	Quiz Attempt	150	3000
firefox	Submission and Result	400	3000
firefox	Attempt History	138	3000
firefox	Quiz View	241	3000
firefox	Quiz Attempt	117	3000
firefox	Submission and Result	367	3000
firefox	Attempt History	187	3000
firefox	Subject Performance	194	3000
firefox	Admin Dashboard	401	3000

Execution observations

Browser journeys and page-load timings use the compiled production frontend with same-origin API proxying and disposable real MongoDB. The 3000 ms assertion is unchanged. Earlier development-server timing failures remain in their original dated pack.

Post-run verification and integration

The previously intermittent Student journey was repeated three times in each browser on the same tested source: 9/9 additional scenarios passed, zero failed/skipped/flaky cases. Maximum content measurements were Chrome 351 ms, Edge 321 ms and Firefox 372 ms, using the unchanged <3,000 ms limit. These are additional rechecks, not replacements for or additions to the 541 planned case count. Per-repeat evidence.

Defined API workload revalidation: Quiz p90 79 ms at 10 sessions and 307 ms at 50 sessions; auth/material smoke p90 53 ms and 250 ms respectively. The earlier 23,471 ms fifty-session Quiz observation was not reproduced; it remains historical evidence, and no universal load-latency guarantee is claimed. Dated follow-up and defect disposition.

PR #44 merged into main at 1c6ecafc3807a1f3916d79ea81416cd580e12eec. The exact final PR-head CI run 41990092049 passed before merge; the application source also passed the linked source-specific CI run. No independent review signature, user approval or Guide sign-off is invented. Manual acceptance register.

Executed browser scenario inventory

Evidence ID	Scenario / installed browser	Result
BROWSER-001	[chrome] bulk material moderation confirms hidden scope, supports cancellation and retains PDF audit data	PASS
BROWSER-002	[chrome] complete Student attempt, result, history and subject performance survive Quiz moderation	PASS
BROWSER-003	[chrome] real Quiz creation, discovery, trusted API scoring and confirmed Quiz moderation integrate	PASS
BROWSER-004	[chrome] student uploads, administrator confirms removal and removed content cannot be downloaded	PASS
BROWSER-005	[chrome] student loads the 26th search match and keeps the search after another upload	PASS
BROWSER-006	[chrome] Admin dashboard integrates real Question Bank moderation, responsive navigation and Student denial	PASS
BROWSER-007	[chrome] ten isolated browser sessions concurrently upload, create quizzes, submit and read their own history	PASS
BROWSER-008	[edge] bulk material moderation confirms hidden scope, supports cancellation and retains PDF audit data	PASS
BROWSER-009	[edge] complete Student attempt, result, history and subject performance survive Quiz moderation	PASS
BROWSER-010	[edge] real Quiz creation, discovery, trusted API scoring and confirmed Quiz moderation integrate	PASS
BROWSER-011	[edge] student uploads, administrator confirms removal and removed content cannot be downloaded	PASS
BROWSER-012	[edge] student loads the 26th search match and keeps the search after another upload	PASS
BROWSER-013	[edge] Admin dashboard integrates real Question Bank moderation, responsive navigation and Student denial	PASS
BROWSER-014	[edge] ten isolated browser sessions concurrently upload, create quizzes, submit and read their own history	PASS
BROWSER-015	[firefox] bulk material moderation confirms hidden scope, supports cancellation and retains PDF audit data	PASS
BROWSER-016	[firefox] complete Student attempt, result, history and subject performance survive Quiz moderation	PASS
BROWSER-017	[firefox] real Quiz creation, discovery, trusted API scoring and confirmed Quiz moderation integrate	PASS
BROWSER-018	[firefox] student uploads, administrator confirms removal and removed content cannot be downloaded	PASS
BROWSER-019	[firefox] student loads the 26th search match and keeps the search after another upload	PASS
BROWSER-020	[firefox] Admin dashboard integrates real Question Bank moderation, responsive navigation and Student denial	PASS
BROWSER-021	[firefox] ten isolated browser sessions concurrently upload, create quizzes, submit and read their own history	PASS